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EXPERIMENT 17

Exp No:17 Experiment to understand Linear Regression for a given data set

AIM:

To understand Linear Regression for a given data set.

DESCRIPTION:

Understand the Linear regression for the dataset given.

CODE:

```
import numpy as np

import pandas as pd

df = pd.read_csv('/content/Salary_Data.csv')

print(df)

df.info()

df.dropna(inplace=True)

df.info()

print(df.describe())

features = df.iloc[:, [4]].values

label = df.iloc[:, [5]].values

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(

    features, label, test_size=0.2, random_state=0)

from sklearn.linear_model import LinearRegression

model = LinearRegression()
```

```
model.fit(x_train, y_train)

print(model.score(x_train, y_train))

print(model.score(x_test, y_test))

print(model.coef_)

print(model.intercept_)

import pickle

pickle.dump(model, open('SalaryPred.model', 'wb'))

model = pickle.load(open('SalaryPred.model', 'rb'))

yr_of_exp = float(input("Enter Years of Experience: "))

yr_of_exp_NP = np.array([[yr_of_exp]])

Salary = model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}".format(yr_of_exp,
Salary[0][0]))
```

OUTPUT:

```
       Age  Gender    Education Level             Job Title  \
0     32.0    Male         Bachelor's      Software Engineer
1     28.0  Female           Master's           Data Analyst
2     45.0    Male                PhD         Senior Manager
3     36.0  Female         Bachelor's        Sales Associate
4     52.0    Male           Master's               Director
...    ...     ...                ...                    ...
6699  49.0  Female                PhD  Director of Marketing
6700  32.0    Male        High School        Sales Associate
6701  30.0  Female  Bachelor's Degree      Financial Manager
6702  46.0    Male    Master's Degree      Marketing Manager
6703  26.0  Female        High School        Sales Executive
```

```
      Years of Experience    Salary
0                     5.0   90000.0
1                     3.0   65000.0
2                    15.0  150000.0
3                     7.0   60000.0
4                    20.0  200000.0
...                   ...       ...
6699                 20.0  200000.0
6700                  3.0   50000.0
6701                  4.0   55000.0
6702                 14.0  140000.0
6703                  1.0   35000.0

[6704 rows x 6 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6704 entries, 0 to 6703
```

```
Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Age                  6702 non-null   float64
 1   Gender               6702 non-null   object
 2   Education Level      6701 non-null   object
 3   Job Title            6702 non-null   object
 4   Years of Experience  6701 non-null   float64
 5   Salary               6699 non-null   float64
dtypes: float64(3), object(3)
memory usage: 314.4+ KB
<class 'pandas.core.frame.DataFrame'>
Index: 6698 entries, 0 to 6703

Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Age                  6698 non-null   float64
 1   Gender               6698 non-null   object
 2   Education Level      6698 non-null   object
 3   Job Title            6698 non-null   object
 4   Years of Experience  6698 non-null   float64
 5   Salary               6698 non-null   float64
dtypes: float64(3), object(3)
memory usage: 366.3+ KB

                Age  Years of Experience         Salary
 count  6698.000000          6698.000000    6698.000000
 mean     33.623022             8.095178  115329.253061
 std       7.615784             6.060291   52789.792507
 min      21.000000             0.000000     350.000000
 25%      28.000000             3.000000   70000.000000
 50%      32.000000             7.000000  115000.000000
 75%      38.000000            12.000000  160000.000000
 max      62.000000            34.000000  250000.000000
 0.6517371248411507
 0.664530135385327
 [[7044.04234834]]
 [58560.39643187]
 Enter Years of Experience: 45
 Estimated Salary for 45.0 years of experience is 375542.30210725486
```

RESULT:

Thus, a Linear Regression model was implemented successfully to predict salary based on years of experience. The model was evaluated using training and testing data, and the trained model was saved using Pickle for future predictions.